

BRISTOL WEATHER CENTRE, United Kingdom

WMO: 037260

Lat: 51.467N

Lon: 2.600W

Elev: 36

StdP: 14.68

Time Zone: 0.00 (EUW)

Period: 90-01

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.5	31.1	20.2	15.2	34.8	23.3	17.6	36.3	28.5	48.1	26.0	49.0	4.8	70	0.532

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.4	80.1	65.0	76.6	63.0	73.4	61.9	66.7	75.8	64.8	72.6	63.2	70.2	7.3	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.5	87.8	69.4	61.9	82.8	67.7	60.4	78.6	66.2	31.3	76.1	29.8	72.8	28.6	70.0	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.9	19.7	17.2	DB	24.8	85.0	2.3	4.6	23.1	88.2	21.7	90.9	20.5	93.5	18.8	96.8	(4)
(5)			WB	23.3	69.4	2.9	1.5	21.2	70.5	19.5	71.4	17.8	72.3	15.7	73.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.5	43.1	43.8	46.8	48.9	55.7	59.8	64.0	64.4	59.1	53.0	47.7	43.1	(6)	
	DBStd	9.09	5.43	5.41	4.26	4.49	5.43	4.45	4.18	4.58	3.70	4.78	5.11	6.27	(7)	
	HDD50	947	219	179	113	72	11	0	0	0	0	26	103	224	(8)	
	HDD65	4670	680	593	564	484	293	169	70	67	180	371	520	679	(9)	
	CDD50	1857	4	6	14	38	186	293	433	446	274	119	33	11	(10)	
	CDD65	106	0	0	0	0	3	12	39	48	4	0	0	0	(11)	
	CDH74	608	0	0	0	0	27	89	210	268	14	0	0	0	(12)	
	CDH80	107	0	0	0	0	0	14	24	69	0	0	0	0	(13)	
(14) Wind	WSAvg	7.3	8.5	8.3	8.0	7.8	7.1	7.1	6.9	6.1	6.0	7.0	6.7	8.0	(14)	
(15) Precipitation	PrecAvg	35.2	3.6	2.4	2.7	2.3	2.4	2.6	2.4	3.0	3.3	3.5	3.6	3.7	(15)	
	PrecMax	44.2	6.9	5.7	6.4	4.9	5.9	6.2	5.9	7.1	7.9	8.8	8.3	8.3	(16)	
	PrecMin	25.0	0.4	0.1	0.2	0.1	0.3	0.5	0.3	0.3	0.8	0.2	1.3	0.8	(17)	
	PrecStd	4.7	1.9	1.5	1.3	1.2	1.4	1.6	1.4	1.7	1.8	2.0	1.7	1.9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.6	58.0	62.1	67.8	77.2	82.1	82.8	87.1	76.5	67.5	60.2	57.2	(19)	
		MCWB	51.5	51.8	52.6	54.0	60.2	66.2	66.7	66.0	64.7	60.5	56.9	54.2	(20)	
	2%	DB	53.8	54.4	57.1	62.1	73.7	76.1	79.6	80.9	71.3	63.5	58.3	55.6	(21)	
		MCWB	50.4	50.4	50.1	52.0	60.1	62.9	65.0	65.0	61.5	57.9	55.4	53.0	(22)	
	5%	DB	52.2	52.2	54.7	58.4	69.7	71.8	76.1	76.0	67.8	61.7	56.7	53.8	(23)	
		MCWB	49.4	48.6	49.2	50.7	58.4	60.1	62.7	63.6	59.3	57.1	53.9	51.3	(24)	
	10%	DB	50.8	50.8	52.8	55.8	65.3	67.9	72.5	72.2	65.6	60.2	55.2	52.1	(25)	
		MCWB	48.1	47.6	48.2	49.6	56.3	58.3	62.1	61.9	58.3	56.1	52.5	49.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.6	53.1	53.9	57.3	63.9	67.8	75.2	68.4	65.5	61.4	57.6	54.8	(27)	
		MCDB	54.6	56.7	59.2	63.6	73.0	79.4	75.2	79.7	73.5	65.5	59.3	56.3	(28)	
	2%	WB	51.2	51.1	52.1	53.6	61.5	63.4	66.6	66.8	63.1	59.6	56.0	53.2	(29)	
		MCDB	53.3	53.6	55.4	58.3	70.3	73.6	75.0	76.1	68.5	62.6	57.7	55.0	(30)	
	5%	WB	49.8	49.5	50.7	52.0	59.2	61.2	64.3	65.2	61.7	58.1	54.5	51.7	(31)	
		MCDB	51.9	51.8	53.3	56.9	67.5	69.2	71.3	73.1	66.0	60.9	56.4	53.5	(32)	
	10%	WB	48.3	48.1	49.5	50.6	57.2	59.4	62.7	63.5	60.1	56.6	52.7	50.0	(33)	
		MCDB	50.5	50.6	51.9	54.7	64.8	66.1	70.0	70.2	64.1	59.7	54.9	51.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.6	8.5	9.5	11.2	13.1	12.5	12.5	12.4	11.0	9.7	8.1	7.3	(35)	
		MCDBR	8.6	9.3	12.1	15.0	20.1	19.6	19.7	18.7	14.6	10.6	7.8	8.0	(36)	
	5% WB	MCWBR	8.1	7.8	7.5	8.2	9.9	9.7	9.6	7.2	7.8	7.3	6.8	7.8	(37)	
		MCWBR	8.5	9.1	9.5	12.7	17.7	17.0	15.6	15.1	12.7	9.6	7.7	8.2	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.330	0.344	0.378	0.399	0.403	0.396	0.393	0.380	0.370	0.356	0.330	0.319	(40)	
		taud	2.349	2.370	2.276	2.236	2.262	2.315	2.330	2.383	2.405	2.416	2.399	2.329	(41)	
	Edn at Noon	Edn at Noon	213	246	257	266	270	272	271	269	259	238	213	196	(42)	
		Edh at Noon	20	26	34	40	41	39	38	34	30	24	19	18	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	248	469	830	1277	1569	1703	1595	1321	995	563	299	193	(44)	
	RadStd	RadStd	23	57	106	145	79	180	150	116	73	49	29	24	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (76 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page